Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/09/2005
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Project Title:	Aggregation and Gelation in Suspensions of Anisometric Particles: Studies of Colloidal Platelets and Rods		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
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Zip Code:	48109-2136	**Congressional District:**	15
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Solomon, Michael (University of Michigan)		
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Task Description:	Overview: This project seeks to understand the effect of anisometric shape on colloidal aggregation and gelation. During this period, five subtasks have been accomplished as described below. The aggregation and gelation of colloidal suspensions are fundamental processes in fluid physics that incorporate elements of statistical mechanics, hydrodynamics, and physical chemistry. Quantifying and predicting the microstructure of colloidal aggregates and gels is a basic scientific task that has been investigated by fluid physicists for more than twenty years. Aggregation and gelation underpin technology relevant to the colloidal processing of ceramics, the development of polymer/particle composites, the stabilization of paints, and the flocculation of solid particulates. While tremendous progress has been made in understanding the aggregation and gelation in suspensions of spherical colloids, the situation for suspensions of anisometric colloids is less impressive. Specifically, the microstructure of aggregates and gels of colloidal platelets and rods is believed to profoundly differ from spherical suspensions.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	I. Effect of monomer aspect ratio on colloidal rod aggregation. Colloidal boehmite rods were dispersed in aqueous medium and aggregation induced by the addition of divalent salt to the initially stable suspension. The fractal structure of clusters formed by diffusion limited aggregation of rod-like particles was characterized over three decades of the scattering vector q. The apparent fractal dimension displayed an unexpected dependence on the aspect ratio of the constituent monomers. Monte Carlo simulations of aggregating Brownian rods corroborated the experimental finding that the measured fractal dimension is an increasing function of the monomer aspect ratio. Moreover, increasing the rod aspect ratio eliminated the structural distinction between diffusion- and reaction-limited cluster aggregation that is observed for spheres. While the role of monomer shape anisotropy in the formation of equilibrium liquid crystal phases has long been known, this task discovered unforeseen implications of this parameter for amorphous, isotropic structures that facilitate a better understanding of the unusual rheology, sedimentation, packing, percolation, and self-assembly of anisometric colloids and nanorods. II. Effect of monomer aspect ratio on the dynamics of colloidal rod gels. The dynamics of gels comprised of rods has been studied by means of dynamic light scattering. The internal dynamics of the gels, as parameterized by the Krall-Weitz model, indicate that short-range motions in the gels depend on rod aspect ratio in a way that was proposed by Philipse. Non-ergodic behavior, characteristic of gel dynamics, was observed at very low volume fractions. The arrested dynamics and gelation at this remarkably low volume fraction is likely due to the large aspect ratio of the colloidal particles from which the clusters are formed. In this way, we observe that the high aspect ratio of the rods facilitates the formation of stable gels at volume fractions that would not be possible for spherical colloids of similar density. III. Effect of shear deformation on the orientational anisotropy of colloidal aggregates. By means of real-time flow light scattering we initiated the quantification of the transient structural orientation and break-up of Solomon diffusion-limited cluster aggregation (DLCA) colloidal gels during start-up of steady shear flow. For these studies, a unique flow/gradient plane Taylor-Couette shear cell was designed and fabricated. The device geometry allows the study of the effect of transient shear flow on rod or sphere aggregate and gel orientation. The measurements for DLCA colloidal gels of spheres, in the flow-gradient plane, indicate the way in which shear flow affects of the structure and orientation of fractal objects. Our results identified three distinct regimes during start-up of steady shear for colloidal DLCA gels: orientation, break-up, and steady-state. The first regime (orientation) exhibited universal scaling with the applied strain, a characteristic of high-Pe phenomena. Moreover, little to no relaxation of the structure was observed upon cessation of flow in this regime. The break-up of the fractal network is recognized by a sharp peak in the normalized degree of anisotropy, which is followed by relaxation of the structure to a steady state. These data with spheres can be used for purposes of comparison with later rod studies. IV. Confocal microscopy of spherical colloidal gels. To more carefully assess the role of sedimentation in anisometric suspensions and to study distributions of structural and dynamical quantities in gels formed from anisometric colloids, we aimed to directly visualize structure and dynamics by means of confocal laser scanning microscopy under quiescent and flow conditions. To obtain data to which the anisometric colloidal studies could be compared, we first synthesized gels consisting of spherical colloids and observed their gel structure by means of confocal laser scanning microscopy (CLSM). We identified long-range density correlations and voids as the dominant structural features of dense colloidal gels. This study used recent developments in fluorescent particle synthesis, CLSM and quantitative imaging processing to identify these new features of gel structure. We discovered that unconstrained uniaxial compression (squeeze flow) induces the formation of voids and cracks in dense colloidal suspensions. Our observations of non-ideal colloidal deformation on the mesoscopic scale of 10-100 particle diameters indicate that explicit consideration of this scale is necessary to understand and model the rheology of dense colloidal gels. V. Preparation of fluorescent ellipsoidal colloidal rods for studies of aggregation and gelation. Stable suspensions of monodisperse fluorescent polymeric microspheroids were synthesized as model systems for the future direct visualization of amorphous and ordered structures in the suspensions of anisometric colloids. The method, adapted from literature references, involves the preparation of monodisperse sterically stabilized Poly(methyl methacrylate) (PMMA) microspheres. These colloids are then subsequently deformed in a uniaxial extensional flow field to generate anisometric particles such as ellipsoids and spherocylinders. The size of these anisometric particles (a ~ 5 microns, b ~ 0.5 microns) makes them suitable for confocal microscopy studies. They can be dispersed in refractive-index matching mixed organic solvents to minimize the effects of scattering when imaging deep into the sample volume. Issues that ongoing work seeks to resolve are the yield of the synthetic scheme and characterization of the interactions between the particles. VI. Outreach activities Our program in colloidal science and rheology created a discovery module in the University of Michigan Materials Science Focus Project-Future Science Future Engineering Program. During the module, partially supported by NASA and executed three times during this grant period, about fifteen middle school girls prepared complex fluids from colloids and polymers, experimented with them and observed their structure by optical and scanning electron microscopy. The complex fluids studied were highly elastic Boger fluids, shear thickening suspensions, colloidal gels and immiscible blends and emulsions. All the fluids were prepared from nontoxic components soluble in aqueous media.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Mohraz A, Moler DB, Ziff RM, Solomon MJ. "Effect of monomer geometry on the fractal structure of colloidal rod aggregates." Phys Rev Lett. 2004 Apr 16;92(15):155503. Epub 2004 Apr 14. PubMed PMID: 15169295 , Apr-2004